

Pavan Bobba

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EXPERIENCE

Georgia State University

Atlanta, GA

Software Engineer

Sep 2024 – Apr 2026

- Built a Proof of Concept (POC) **NLQ-to-SQL agent** using LangChain and AWS Bedrock to evaluate natural-language access to **healthcare data**, generating secure SQL queries and reducing ad-hoc reporting tickets by **60%**.
- Built a **RAG-based** assistant to answer student health policy and insurance coverage questions across **200+ scenarios**, improving access to accurate plan, eligibility, and claims information.
- Developed a full-stack student management platform using **React.js, TypeScript, Node.js, and AWS DynamoDB**, automating administrative workflows for 300+ professors and 10K+ students while reducing manual workload by **85%** and saving **20+ hours weekly**.
- Designed an event-driven notification pipeline where **DynamoDB Streams** triggered **AWS Lambda** to send application status, enrollment, and deadline alerts through the SendGrid API, reaching 10K+ students at a **98%** delivery rate.

DBS Bank

Hyderabad, IN

Software Engineer I (Full-Stack Engineer)

Sep 2023 - Jul 2024

- Cut mobile analytics event loss from **2.5% to below 0.3%** across 200K daily events by rebuilding **Kafka pipelines** with at-least-once delivery, bounded retries, and consumer-group scaling, which reduced **peak lag by 65%**.
- Delivered 99.9%+ API availability for UnionPay QR payments across 15 markets at **100K+ requests/day** by developing Spring Boot microservices with merchant validation, multi-currency routing, and fault-tolerant transaction processing.
- Evaluated deduplication approaches for Kafka retries and implemented persistent idempotency records with atomic checks, eliminating **95%+ of duplicate mobile events** during retries and restarts..
- Reduced incident detection time by **70%** by building a **Log4j** and **AWS CloudWatch** monitoring stack, while increasing test coverage to 95%+ through standardized JUnit and Mockito testing practices.

Junior Software Engineer

Jul 2022 - Sep 2023

- Created reusable **event models and payload schemas** across 4+ customer journeys, reducing new-journey instrumentation effort by **50%** through standardized event structures.
- Engineered **REST ingestion APIs** using **Spring Boot and Hibernate ORM** to process 150K+ mobile behavioral records/day for downstream product analytics pipelines.
- Implemented **validation and normalization layers** within the ingestion service to reject malformed events, standardize inconsistent fields, and capture processing failures, reducing downstream data-cleanup effort by **30%**.
- Reduced weekly operations report runtime from **20 minutes to under 3 minutes** by rewriting nested **subqueries and multi-table joins** as CTEs and window functions backed by a covering index, moving exception review from next-day to same-morning.

Tech Neon Solution Private Limited

Hyderabad, IN

Software Engineer

Jun 2021 - May 2022

- Built **4+ stateless microservices** with **REST APIs** to process customer and transaction data, cutting average integration time for new service onboarding by 30%
- Deployed **serverless data workflows** with **AWS Lambda and S3** to automate file uploads, CSV extraction, and downstream processing, replacing manual handling for 10K+ records per
- Reduced nightly healthcare claims aggregation time by **30%** by supporting data ingestion from **Oracle and Microsoft SQL Server** into HDFS using Sqoop and Apache Hive.

SKILLS

- Programming:** Java, Python, JavaScript, TypeScript, SQL
- Backend & APIs:** Spring Boot, Node.js, Express.js, FastAPI, REST APIs, GraphQL, gRPC, Hibernate ORM, JPA
- Databases & Messaging:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Apache Kafka, RabbitMQ
- Cloud, DevOps & Observability:** AWS, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Grafana, Prometheus
- AI/ML Engineering:** LangChain, LangGraph, AI Agents, MCP, Hugging Face, ChromaDB, FAISS, PyTorch
- Certifications:** AWS Certified AI Practitioner, AWS Certified Solutions Architect – Associate

PROJECTS

PolicyPilot | [GitHub](#) | [Live Demo](#)

Next.js, TypeScript, PostgreSQL, GitHub Actions, Terraform

- Developed a **full-stack health insurance advisor** with **Python, LangGraph, Streamlit, and ChromaDB**, combining document retrieval and **LLM reasoning** behind a conversational interface that answered coverage, eligibility, plan, and claims questions.
- Integrated Groq and Google Gemini models with CMS.gov and VA.gov **data sources, creating retrieval, verification, and response-generation workflows** that improved answer grounding across multi-turn insurance conversation

StockFlow | [GitHub](#) | [Live Demo](#)

Python, FastAPI, LangGraph, LangChain, MCP, PostGIS, SQLAlchemy, Docker, Prometheus

- Designed a **LangGraph multi-agent system** for franchise inventory that coordinates forecasting, replenishment, transfer, and approval agents through deterministic tool calls, persisted reasoning traces, and PostGIS-based transfer logic.
- Exposed **11 tools, 4 resources, and 3 prompts through an MCP adapter** over stdio and HTTP JSON-RPC, letting external AI assistants inspect supply-chain state and run simulator actions.

EDUCATION

Georgia State University

Atlanta, GA

Master of Science in Computer Science - GPA:3.80

Aug 2024 - May 2026